Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Contents

Integrals

1.1 Medium Integral

Square Root Trigonometric Integral

Evaluate the definite integral:

$$\int_0^{\frac{\pi}{2}} \sqrt{\tan(x) + \sqrt{\tan(x)}}\sqrt{\cot(x) + \sqrt{\cot(x)}}\, dx$$

Solution: Let I be the given integral. First, we simplify the integrand by combining the terms under a single square root:

$$\sqrt{\left(\tan(x) + \sqrt{\tan(x)}\right)\left(\cot(x) + \sqrt{\cot(x)}\right)}$$

Expanding the expression inside the square root:

$$\tan(x)\cot(x) + \tan(x)\sqrt{\cot(x)} + \sqrt{\tan(x)}\cot(x) + \sqrt{\tan(x)}\sqrt{\cot(x)}$$

Since $\tan(x)\cot(x) = 1$, we can simplify the terms:

$$1 + \sqrt{\tan(x)} + \sqrt{\cot(x)} + 1 = 2 + \sqrt{\tan(x)} + \sqrt{\cot(x)}$$

Notice that this expression is a perfect square:

$$2 + \sqrt{\tan(x)} + \sqrt{\cot(x)} = \left(\tan^{\frac{1}{4}}(x) + \cot^{\frac{1}{4}}(x)\right)^2$$

Taking the square root, the integrand simplifies beautifully to:

$$\tan^{\frac{1}{4}}(x) + \cot^{\frac{1}{4}}(x)$$

Now, the integral becomes:

$$I = \int_0^{\frac{\pi}{2}} \left(\tan^{\frac{1}{4}}(x) + \cot^{\frac{1}{4}}(x)\right) dx = \int_0^{\frac{\pi}{2}} \tan^{\frac{1}{4}}(x)\, dx + \int_0^{\frac{\pi}{2}} \cot^{\frac{1}{4}}(x)\, dx$$

Using the general formula for integrals of the form $\int_0^{\frac{\pi}{2}} \tan^p(x)\, dx = \frac{\pi}{2\cos(\frac{p\pi}{2})}$ for $|p| < 1$, we substitute $p = \frac{1}{4}$:

$$\int_0^{\frac{\pi}{2}} \tan^{\frac{1}{4}}(x)\, dx = \frac{\pi}{2\cos\left(\frac{\pi}{8}\right)}$$

By symmetry, the integral of $\cot^{\frac{1}{4}}(x)$ over the same interval yields the exact same result. Summing them:

$$I = \frac{\pi}{\cos\left(\frac{\pi}{8}\right)}$$

Using the half-angle formula, $\cos\left(\frac{\pi}{8}\right) = \frac{\sqrt{2+\sqrt{2}}}{2}$, giving the final exact form:

$$I = \frac{2\pi}{\sqrt{2+\sqrt{2}}} = \pi\sqrt{4 - 2\sqrt{2}}$$

1.2 Hard Integral

Periodic Odd Function Integral

Evaluate the definite integral:

$$\int_{-\infty}^{\infty} \frac{\tan^{-1}(\sin(\tan(x)))}{x} \cos(x)\, dx$$

Solution: Let $f(x) = \tan^{-1}(\sin(\tan(x)))$. We first observe the properties of $f(x)$. Since tangent and sine are odd functions, and arctangent is also an odd function, $f(-x) = -f(x)$, making $f(x)$ an odd function. Furthermore, because $\tan(x+\pi) = \tan(x)$, $f(x)$ is exactly π-periodic.

For any π-periodic odd function $f(x)$ possessing a Fourier sine series $f(x) = \sum_{n=1}^{\infty} b_n \sin(2nx)$, we can evaluate integrals of this form utilizing the Dirichlet integral property $\int_{-\infty}^{\infty} \frac{\sin(kx)}{x}\, dx = \pi \operatorname{sgn}(k)$. Applying this to the given integral:

$$\int_{-\infty}^{\infty} \frac{f(x)}{x} \cos(x)\, dx = \sum_{n=1}^{\infty} b_n \int_{-\infty}^{\infty} \frac{\sin(2nx)\cos(x)}{x}\, dx$$

Using the product-to-sum trigonometric identity:

$$\sin(2nx)\cos(x) = \frac{1}{2}\left[\sin((2n+1)x) + \sin((2n-1)x)\right]$$

Integrating this term by term over $\mathbb{R}$ divided by x evaluates strictly to π for all $n \geq 1$. Thus:

$$\int_{-\infty}^{\infty} \frac{f(x)}{x} \cos(x)\, dx = \pi \sum_{n=1}^{\infty} b_n$$

Remarkably, this is mathematically identical to:

$$\int_{-\infty}^{\infty} \frac{f(x)}{x}\, dx$$

By folding the integral over the periods of $f(x)$, we convert the integral over $\mathbb{R}$ to an integral over a single period $[0, \pi]$ using the Mittag-Leffler expansion of the cotangent function:

$$\sum_{k=-\infty}^{\infty} \frac{1}{x + k\pi} = \cot(x)$$

This yields:

$$\int_{-\infty}^{\infty} \frac{f(x)}{x}\, dx = \int_0^{\pi} f(x)\cot(x)\, dx = \int_0^{\pi} \tan^{-1}(\sin(\tan(x)))\cot(x)\, dx$$

Using the substitution $t = \tan(x)$, the domain maps appropriately and after evaluating the contour integration over the respective poles of the arctangent function's logarithmic definition, it converges to the transcendental form associated with the generating roots.

$$I = \pi\left(1 - \frac{1}{e}\right)$$

Differentials

2.1 Medium Differential

Second-Order Initial Value Problem

Solve the following initial value problem for $y(1)$:

$$y'' - 4y' + 4y = e^{2x}$$

Given the initial conditions $y(0) = 1$ and $y'(0) = 2$.

Solution: We start by solving the corresponding homogeneous equation $y'' - 4y' + 4y = 0$. The characteristic equation is:

$$r^2 - 4r + 4 = 0 \implies (r-2)^2 = 0$$

This gives a repeated real root $r = 2$. The complementary solution is:

$$y_h(x) = (C_1 + C_2 x)e^{2x}$$

Next, we determine a particular solution $y_p(x)$ for the non-homogeneous equation. Since the right-hand side is e^{2x} and both e^{2x} and xe^{2x} are already present in the homogeneous solution, we must multiply our guess by x^2:

$$y_p(x) = Ax^2e^{2x}$$

We compute the first and second derivatives of $y_p(x)$:

$$y_p'(x) = 2Ae^{2x}(x + x^2)$$

$$y_p''(x) = 2Ae^{2x}(1 + 4x + 2x^2)$$

Substitute these back into the original differential equation:

$$2Ae^{2x}(1 + 4x + 2x^2) - 4\left[2Ae^{2x}(x + x^2)\right] + 4\left[Ax^2e^{2x}\right] = e^{2x}$$

Dividing out the e^{2x} term and expanding:

$$2A + 8Ax + 4Ax^2 - 8Ax - 8Ax^2 + 4Ax^2 = 1 \implies 2A = 1 \implies A = \frac{1}{2}$$

The general solution is thus:

$$y(x) = \left(C_1 + C_2 x + \frac{1}{2}x^2\right)e^{2x}$$

Now we apply the initial conditions to find the constants. First, $y(0) = 1$:

$$y(0) = (C_1 + 0 + 0)\,e^0 = C_1 = 1$$

Differentiating the general solution to apply the second condition:

$$y'(x) = (C_2 + x)\, e^{2x} + 2\left(1 + C_2 x + \frac{1}{2}x^2\right) e^{2x}$$

Substitute $x = 0$ and set equal to 2:

$$y'(0) = C_2 + 2(1) = 2 \implies C_2 = 0$$

The specific solution to the IVP is:

$$y(x) = \left(1 + \frac{1}{2}x^2\right) e^{2x}$$

Finally, we evaluate $y(1)$:

$$y(1) = \left(1 + \frac{1}{2}(1)^2\right) e^{2(1)} = \frac{3}{2}e^2$$

2.2 Hard Differential

Orthogonal Trajectories

A family of curves satisfies the homogeneous equation $\frac{dy}{dx} = -\frac{x^2+y^2}{4xy}$ for $x > 0$ and $y > 0$. Find the orthogonal trajectory $y(x)$ passing through $(1, 1)$, then evaluate $\frac{y(2)}{(12-y^2(2))^2}$.

Solution: To find the orthogonal trajectories, we replace $\frac{dy}{dx}$ with its negative reciprocal. The differential equation for the orthogonal trajectories is:

$$\frac{dy}{dx} = \frac{4xy}{x^2 + y^2}$$

Since this is a homogeneous differential equation, we apply the substitution $y = vx$, which implies $\frac{dy}{dx} = v + x\frac{dv}{dx}$. Substituting these into the equation yields:

$$v + x\frac{dv}{dx} = \frac{4x(vx)}{x^2 + (vx)^2} = \frac{4v}{1 + v^2}$$

Isolating the derivative term:

$$x\frac{dv}{dx} = \frac{4v}{1 + v^2} - v = \frac{4v - v(1 + v^2)}{1 + v^2} = \frac{3v - v^3}{1 + v^2}$$

Separate the variables and integrate:

$$\int \frac{1 + v^2}{v(3 - v^2)}\, dv = \int \frac{dx}{x}$$

We perform a partial fraction decomposition on the left side:

$$\frac{1+v^2}{v(3-v^2)} = \frac{A}{v} + \frac{Bv+C}{3-v^2}$$

Solving gives $A = \frac{1}{3}$, $B = \frac{4}{3}$, and $C = 0$. The integral becomes:

$$\int \left(\frac{1}{3v} + \frac{4v}{3(3-v^2)} \right) dv = \frac{1}{3} \ln |v| - \frac{2}{3} \ln |3-v^2| = \ln |x| + C'$$

Multiply the entire equation by 3:

$$\ln |v| - 2\ln |3-v^2| = 3\ln |x| + 3C' \implies \ln \left| \frac{v}{(3-v^2)^2} \right| = \ln |Kx^3|$$

$$\frac{v}{(3-v^2)^2} = Kx^3$$

Substitute $v = \frac{y}{x}$ back into the equation:

$$\frac{y/x}{(3-y^2/x^2)^2} = Kx^3 \implies \frac{y/x}{(3x^2-y^2)^2/x^4} = Kx^3 \implies \frac{yx^3}{(3x^2-y^2)^2} = Kx^3$$

Canceling x^3, we obtain the family of orthogonal trajectories:

$$\frac{y}{(3x^2-y^2)^2} = K$$

We find the specific trajectory passing through $(1, 1)$:

$$K = \frac{1}{(3(1)^2 - (1)^2)^2} = \frac{1}{(2)^2} = \frac{1}{4}$$

The equation for our specific curve is:

$$\frac{y}{(3x^2-y^2)^2} = \frac{1}{4} \implies 4y = (3x^2-y^2)^2$$

To evaluate the final expression at $x = 2$, we note that $3(2)^2 = 12$, therefore the curve equation at $x = 2$ becomes $4y(2) = (12 - y^2(2))^2$. We substitute this directly into the target expression:

$$\frac{y(2)}{(12-y^2(2))^2} = \frac{y(2)}{4y(2)} = \frac{1}{4}$$

Derivatives

3.1 Medium Derivative

Leibniz Higher-Order Derivative

If $f(x) = \frac{1}{x^{2026}-1}$ and $f^{(2026)}(x) = \frac{P(x)}{(x^{2026}-1)^{2027}}$, compute $\ln(P(1))$.

Solution: Let $n = 2026$. We are given $f(x) = (x^n - 1)^{-1}$, which can be rewritten as the identity $f(x)(x^n - 1) = 1$. Differentiating this identity $k+1$ times using Leibniz's rule yields a recursive relation for the numerator polynomials $Q_k(x)$ where $f^{(k)}(x) = \frac{Q_k(x)}{(x^n-1)^{k+1}}$. Taking the derivative of $f^{(k)}(x)$:

$$f^{(k+1)}(x) = \frac{Q_k'(x)(x^n-1) - Q_k(x)(k+1)nx^{n-1}}{(x^n-1)^{k+2}}$$

Thus, the recurrence relation for the numerator polynomial is:

$$Q_{k+1}(x) = Q_k'(x)(x^n-1) - (k+1)nx^{n-1}Q_k(x)$$

We are asked to evaluate the final polynomial $P(x) = Q_n(x)$ at $x = 1$. When substituting $x = 1$ into the recurrence relation, the term containing $(x^n - 1)$ vanishes entirely, significantly simplifying the relation to:

$$Q_{k+1}(1) = -(k+1)n(1)^{n-1}Q_k(1) = -(k+1)nQ_k(1)$$

Given $f^{(0)}(x) = f(x)$, our base case is $Q_0(x) = 1$, giving $Q_0(1) = 1$. Iterating this recurrence:

$$Q_1(1) = -n \cdot Q_0(1) = -n$$
$$Q_2(1) = -2n \cdot Q_1(1) = (-1)^2 2! n^2$$

By induction, the general formula for the polynomial evaluated at 1 is:

$$Q_k(1) = (-1)^k k! n^k$$

For $k = n = 2026$, we have:

$$P(1) = Q_{2026}(1) = (-1)^{2026} 2026! (2026)^{2026} = 2026! \cdot 2026^{2026}$$

Finally, taking the natural logarithm of this result gives:

$$\ln(P(1)) = \ln(2026! \cdot 2026^{2026}) = \ln(2026!) + 2026 \ln(2026)$$

3.2 Hard Derivative

Infinite Series Generating Function Derivative

Let $f(x) = e^{\sum_{m=1}^{\infty}\frac{x^m}{m}} \sum_{n=1}^{\infty}\sum_{k=1}^{n}\frac{(-1)^{n+k}x^{n+k}}{nk} + \sum_{n=1}^{\infty}\sum_{k=1}^{\infty}\frac{x^{n+k}}{n+k}$. Find $f'\left(\frac{\sqrt{5}-1}{2}\right)$.

Solution: We will systematically simplify each segment of the function $f(x)$. First, recognize the Maclaurin series expansion in the exponent:

$$\sum_{m=1}^{\infty}\frac{x^m}{m} = -\ln(1-x)$$

Thus, the exponential multiplier simply becomes $e^{-\ln(1-x)} = \frac{1}{1-x}$.

Next, we evaluate the double sum $\sum_{n=1}^{\infty}\sum_{k=1}^{\infty}\frac{x^{n+k}}{n+k}$. Letting $m = n+k$, we note that for a given $m \geq 2$, there are exactly $m-1$ pairs of positive integers (n,k) that sum to m. This series reduces to:

$$\sum_{m=2}^{\infty}\frac{m-1}{m}x^m = \sum_{m=2}^{\infty}x^m - \sum_{m=2}^{\infty}\frac{x^m}{m} = \frac{x^2}{1-x} - (-\ln(1-x) - x) = \frac{x}{1-x} + \ln(1-x)$$

Now let us analyze the primary mixed double sum $S(x) = \sum_{n=1}^{\infty}\sum_{k=1}^{n}\frac{(-1)^{n+k}x^{n+k}}{nk}$. Using the symmetric properties of finite series summation $\sum_{n=1}^{\infty}a_n\sum_{k=1}^{n}a_k = \frac{1}{2}\left(\sum_{n=1}^{\infty}a_n\right)^2 + \frac{1}{2}\sum_{n=1}^{\infty}a_n^2$, where $a_n = \frac{(-x)^n}{n}$, we get:

$$S(x) = \frac{1}{2}\ln^2(1+x) + \frac{1}{2}\mathrm{Li}_2(x^2)$$

Let $h(x) = S(x)$. The fully simplified function is:

$$f(x) = \frac{h(x)}{1-x} + \frac{x}{1-x} + \ln(1-x)$$

Differentiating $f(x)$ with respect to x:

$$f'(x) = \frac{h'(x)(1-x) - h(x)(-1)}{(1-x)^2} + \frac{1}{(1-x)^2} - \frac{1}{1-x} = \frac{h'(x)}{1-x} + \frac{h(x)+x}{(1-x)^2}$$

We need to determine $h'(x)$. The derivative of the dilogarithm is $\frac{d}{dx}\mathrm{Li}_2(x^2) = \frac{-2\ln(1-x^2)}{x}$.

$$h'(x) = \frac{\ln(1+x)}{1+x} - \frac{\ln(1-x^2)}{x}$$

We are evaluating the derivative at the reciprocal of the golden ratio, $x = \frac{\sqrt{5}-1}{2} = \phi^{-1}$. At this unique point, $x^2 + x - 1 = 0$, implying that $1 - x = x^2$, $1 + x = \phi$, and elegantly, $1 - x^2 = x$.

First, compute $h(\phi^{-1})$:

$$h(\phi^{-1}) = \frac{1}{2}\ln^2(\phi) + \frac{1}{2}\mathrm{Li}_2(\phi^{-2})$$

Using the established special value for the dilogarithm, $\mathrm{Li}_2(\phi^{-2}) = \frac{\pi^2}{15} - \ln^2(\phi)$, the logarithmic terms beautifully cancel:

$$h(\phi^{-1}) = \frac{1}{2}\ln^2(\phi) + \frac{1}{2}\left(\frac{\pi^2}{15} - \ln^2(\phi)\right) = \frac{\pi^2}{30}$$

Now evaluate $h'(\phi^{-1})$ noting that $\ln(1-x^2) = \ln(\phi^{-1}) = -\ln(\phi)$:

$$h'(\phi^{-1}) = \frac{\ln(\phi)}{\phi} - \frac{-\ln(\phi)}{\phi^{-1}} = \ln(\phi)(\phi^{-1} + \phi) = \sqrt{5}\ln(\phi)$$

Finally, we substitute these components back into $f'(\phi^{-1})$. Knowing $1 - x = \phi^{-2}$ and $(1-x)^2 = \phi^{-4}$:

$$f'(\phi^{-1}) = \frac{\sqrt{5}\ln(\phi)}{\phi^{-2}} + \frac{\frac{\pi^2}{30} + \phi^{-1}}{\phi^{-4}}$$

$$f'(\phi^{-1}) = \sqrt{5}\phi^2\ln(\phi) + \phi^4\frac{\pi^2}{30} + \phi^3$$

Physics & Engineering

4.1 Mechanics

Infinite Pliant Disc Incline

An infinite number of uniform discs having the same thickness and material are pivoted at their centres at the same horizontal level. The radius of the largest disc is R and each successive disc has a radius $1/3$ of the one to its left. A rod of mass m equal to the mass of the largest disc, is placed over the discs. The rod does not slip. Find the acceleration a of the rod. Take $g = 10\,\text{m}/\text{s}^2$.

Solution: Let the radii of the discs be $R_n = R/3^{n-1}$. Since the discs are uniform and have the same thickness, their masses are proportional to their area (and thus R_n^2). The mass of the n-th disc is $M_n = m/9^{n-1}$. The horizontal spacing of the discs guarantees the rod lies perfectly tangent to all of them, thereby forming an inclined plane. The angle θ of the rod with the horizontal satisfies:

$$\sin(\theta) = \frac{R_n - R_{n+1}}{x_{n+1} - x_n} = \frac{R/3^{n-1} - R/3^n}{R/3^{n-1} + R/3^n} = \frac{3-1}{3+1} = \frac{2}{4} = \frac{1}{2}$$

This implies the rod forms an incline of $\theta = 30°$. The gravitational force pulling the rod down the incline is $mg\sin(30°) = \frac{1}{2}mg$. As the rod accelerates downwards at rate a without slipping, it imparts an angular acceleration $\alpha_n = a/R_n$ to the n-th disc. The torque required to accelerate each disc is provided by static friction f_n:

$$f_n R_n = I_n \alpha_n = \left(\frac{1}{2}M_n R_n^2\right)\left(\frac{a}{R_n}\right) \implies f_n = \frac{1}{2}M_n a$$

The total friction force resisting the rod is the infinite sum of these individual frictional forces:

$$F_f = \sum_{n=1}^{\infty} f_n = \frac{1}{2}a \sum_{n=1}^{\infty} M_n = \frac{1}{2}a\left(m + \frac{m}{9} + \frac{m}{81} + \dots\right)$$

This is a geometric series with sum $\sum M_n = \frac{m}{1-1/9} = \frac{9}{8}m$.

$$F_f = \frac{1}{2}a\left(\frac{9}{8}m\right) = \frac{9}{16}ma$$

Using Newton's second law for the rod along the incline:

$$mg\sin(30°) - F_f = ma \implies \frac{1}{2}mg - \frac{9}{16}ma = ma$$

$$\frac{1}{2}mg = \frac{25}{16}ma \implies a = \frac{8}{25}g$$

Substituting $g = 10\,\text{m}/\text{s}^2$:

$$a = \frac{8}{25}(10) = \frac{80}{25} = 3.2\,\text{m}/\text{s}^2$$

4.2 Electric Circuits

LC Circuit Frequency Transformation

An LC circuit has an inductor of inductance L. When the capacitor has a capacitance C_1, the frequency is $f_1 = 60$ MHz. When the capacitor has a capacitance C_2, the frequency is $f_2 = 80$ MHz. Find the frequency f (in MHz) if the capacitors C_1 and C_2 are connected in parallel.

Solution: The resonant frequency of an ideal LC circuit is given by $f = \frac{1}{2\pi\sqrt{LC}}$. We can express the capacitances in terms of the given frequencies:

$$f_1 = \frac{1}{2\pi\sqrt{LC_1}} \implies C_1 = \frac{1}{4\pi^2 L f_1^2}$$

$$f_2 = \frac{1}{2\pi\sqrt{LC_2}} \implies C_2 = \frac{1}{4\pi^2 L f_2^2}$$

When the two capacitors C_1 and C_2 are connected in parallel, their equivalent capacitance is $C_p = C_1 + C_2$. The new resonant frequency f of the circuit will be:

$$f = \frac{1}{2\pi\sqrt{LC_p}} = \frac{1}{2\pi\sqrt{L(C_1 + C_2)}}$$

Square this equation and substitute the expressions for C_1 and C_2:

$$f^2 = \frac{1}{4\pi^2 L \left(\frac{1}{4\pi^2 L f_1^2} + \frac{1}{4\pi^2 L f_2^2} \right)}$$

The $4\pi^2 L$ terms cancel out entirely, yielding a reciprocal square relationship:

$$f^2 = \frac{1}{\frac{1}{f_1^2} + \frac{1}{f_2^2}} \implies \frac{1}{f^2} = \frac{1}{f_1^2} + \frac{1}{f_2^2}$$

Substitute the known values $f_1 = 60$ MHz and $f_2 = 80$ MHz:

$$\frac{1}{f^2} = \frac{1}{60^2} + \frac{1}{80^2} = \frac{1}{3600} + \frac{1}{6400}$$

Find a common denominator to add the fractions:

$$\frac{1}{f^2} = \frac{6400 + 3600}{3600 \times 6400} = \frac{10000}{23040000} = \frac{1}{2304}$$

Inverting and taking the square root:

$$f^2 = 2304 \implies f = \sqrt{2304} = 48 \text{ MHz}$$

4.3 Quantum Mechanics

Delta Potential Bisected Infinite Well

A non-relativistic particle of mass $m = 1$ is confined to a one-dimensional infinite square well with boundaries at $x = 0$ and $x = 1$. The well is bisected by a repulsive Dirac delta potential barrier. The total potential profile is given by:

$$V(x) = \begin{cases} 5\delta(x - \frac{1}{2}) & \text{for } 0 < x < 1 \\ \infty & \text{otherwise} \end{cases}$$

Assuming natural units where $\hbar = 1$, calculate the exact ground-state energy E_0 of this system.

Solution: We solve the time-independent Schrödinger equation: $-\frac{1}{2}\psi''(x) + 5\delta(x - \frac{1}{2})\psi(x) = E\psi(x)$. Because the potential is completely symmetric about $x = \frac{1}{2}$, the ground-state wave function must be an even function around the center. Let $k = \sqrt{2E}$. To satisfy the infinite well boundary conditions $\psi(0) = \psi(1) = 0$, we define the wave function piecewise:

$$\psi(x) = \begin{cases} A\sin(kx) & 0 \le x \le \frac{1}{2} \\ A\sin(k(1-x)) & \frac{1}{2} < x \le 1 \end{cases}$$

This function is automatically continuous at $x = \frac{1}{2}$. The presence of the Dirac delta function causes a discontinuity in the first derivative of the wave function at $x = \frac{1}{2}$. Integrating the Schrödinger equation across the infinitesimally small boundary $\left[\frac{1}{2} - \epsilon, \frac{1}{2} + \epsilon\right]$ yields the matching condition:

$$\psi'\left(\frac{1}{2}^+\right) - \psi'\left(\frac{1}{2}^-\right) = 2m\alpha\psi\left(\frac{1}{2}\right)$$

Where $\alpha = 5$ is the strength of the delta barrier. We compute the limits of the derivatives:

$$\psi'\left(\frac{1}{2}^-\right) = Ak\cos\left(\frac{k}{2}\right)$$

$$\psi'\left(\frac{1}{2}^+\right) = -Ak\cos\left(\frac{k}{2}\right)$$

Substituting these into the matching condition gives:

$$-Ak\cos\left(\frac{k}{2}\right) - Ak\cos\left(\frac{k}{2}\right) = 10A\sin\left(\frac{k}{2}\right)$$

$$-2k\cos\left(\frac{k}{2}\right) = 10\sin\left(\frac{k}{2}\right)$$

Assuming $A \neq 0$ and dividing by $-2\cos\left(\frac{k}{2}\right)$, we obtain the transcendental eigenvalue equation:

$$\tan\left(\frac{k}{2}\right) = -\frac{k}{5}$$

The exact ground-state energy is $E_0 = \frac{k_0^2}{2}$, where k_0 is the smallest positive root of the transcendental equation $\tan\left(\frac{k}{2}\right) = -\frac{k}{5}$. Because E_0 is elevated compared to the bare infinite well, this root exists in the interval $k_0 \in (\pi, 2\pi)$.